

Roll No. ....

**TPE-001**

**B. TECH. (EC) (FIFTH SEMESTER)**  
**MID SEMESTER EXAMINATION, Oct., 2022**

**ANTENNA AND WAVE PROPAGATION**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain radiation pattern. Name the three different kinds of radiation pattern and draw these radiation patterns in elevation and azimuth plane. Explain HPBW, FNBW, directivity, gain and E-plane. 10 Marks (CO1)

**OR**

- (b) Radiation characteristic of an antenna is as follows :

$$U(\theta, \varphi) = P_m \sin \theta \sin^2 \varphi$$

$$0 \leq \theta \leq \pi, \quad 0 \leq \varphi \leq \pi$$

Determine the exact directivity and compare it with approximate directivity. 10 Marks (CO1)

**P. T. O.**

2. (a) What do you understand by the term efficiency of the antenna ? What are the different types of efficiencies in case of antenna ? Also define the polarization loss factor with *one* example. 10 Marks (CO1)

OR

- (b) Show that  $\nabla^2 \vec{A} + k^2 \vec{A} = -\mu \vec{J}$ , where  $A$  is a magnetic vector potential,  $k$  is wave number,  $\mu$  is permeability and  $J$  is electric current density. 10 Marks (CO1)

3. (a) Explain the reciprocity and duality theorem with proper examples. 10 Marks (CO1, CO2)

OR

- (b) Determine the HPBW and FNBW in elevation and azimuth plane for given radiation intensity. 10 Marks (CO1, CO2)

$$U(\theta, \phi) = 20 \sin^2 \theta \cos 3\phi$$

4. (a) Determine the state of polarization. If circular or elliptical, also determine the sense of rotation, i.e., left hand or right hand.

$$\vec{E} = 5(\hat{a}_x + j\hat{a}_y)e^{-j\beta z} \text{ V/m}$$

If the above polarization state defines the polarization of transmitting antenna and the incident wave to this antenna is RHCP (incident wave will be travelling in negative  $z$ -direction), determine the polarization loss factor. 10 Marks (CO2)

(3)

OR

- (b) An antenna with a gain of 12 dB and an effective aperture of  $3 \text{ m}^2$  is used to receive electromagnetic energy propagating in free space. If the received power is 0.1 nW, what is the power density of the wave at the antenna is replaced by another antenna having a gain of 20 dB, calculate the received power. 10 Marks (CO2)

5. (a) Find out the relationship between antenna effective aperture ( $A_{em}$ ) and directivity (D). 10 Marks (CO2)

OR

- (b) Find out the far field parameters (E- and H-field) for an infinitesimal dipole when the current in z-direction is given as  $I(z') = \hat{a}_z I_o$ , where  $I_o$  is constant. 10 Marks (CO2)